

Torus $1 \rightarrow n$ Amplitudes in the $c = 1$ String

A numerical worldsheet note

25 August 2026

Abstract

This note records the current numerical worldsheet results for torus $1 \rightarrow n$ tachyon amplitudes in the $c = 1$ string. The genus-one $1 \rightarrow 1$ reflection amplitude is evaluated at fifty points on the imaginary-energy ray $\omega = it$, $0 < t < 1$. Its torus two-point block uses necklace internal-weight recursion away from collision, OPE central-charge recursion near collision, and exact finite-level descendant sewing as an audited fallback. The native worldsheet data are frozen before the BRY convention map and matrix-model comparison are applied. A correlated fifty-point fit gives

$$(a, b, c) = (1.0016588, 1.0053852, 1.0093253), \quad \kappa = 1.0082693,$$

with shared-scramble RQMC uncertainties stated below. The genus-one $1 \rightarrow 2$ amplitude is now represented only by its current target-blind ten-point equal-split scan in the pole-free strip. It uses order-eight momentum quadrature, an h -recursion-dominant channel atlas with necklace orders $(8, 3, 3)$, and c -recursion only in the small OPE patches. After the worldsheet scan and its shape analysis were separately frozen, all ten points were found within one quoted RQMC standard error of the BRY-normalized matrix-model curve. No claim at general multiplicity is inferred.

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1 Scope and status

There are two numerical claims and one explicit limitation in this note.

1. The torus $1 \rightarrow 1$ amplitude has a frozen fifty-point worldsheet scan with $0.02 \leq t \leq 0.99$. All points use one common hybrid-recursion, momentum-quadrature, cutoff, tail, and shared-scramble design. The complete native dataset was frozen before the BRY notation map or matrix-model comparison was applied. Sections 3–6 contain only this current result.
2. The torus $1 \rightarrow 2$ amplitude has a frozen target-blind ten-point scan at equal split and $0.05 \leq t \leq 0.95$. Every point uses the same momentum-order-eight, h -recursion-dominant channel atlas and four aligned RQMC scrambles. The bulk region and the direct importance-sampled tail to $\tau_2 = \infty$ are integrated in one run. The worldsheet shape analysis was frozen before the BRY-normalized matrix curve was evaluated.
3. No general torus $1 \rightarrow n$ formula is inferred from either calculation. The two-point and three-point moduli integrals have different dimensions, block structures, and numerical systematics.

The order of operations is fixed throughout: analytically continue to the convergent imaginary-energy chamber, evaluate and serialize the worldsheet integral without importing a target polynomial, freeze the native artifacts, and only then transform conventions or compare with target-space data. The production artifacts record this trust boundary explicitly.

2 Worldsheet and BRY conventions

From this point onward, g_s denotes the BRY string coupling. Thus

$$\boxed{\mu^{-1} = 2\pi g_s.} \tag{1}$$

The older string-note coupling is not used below; when comparison with a legacy artifact is unavoidable it is related by $g_s = 2g_s^{\text{SN}}$.

For the torus two-point function, the target-free sewing normalization is

$$\boxed{\mathcal{A}_1^{\text{ws}}(\omega) = 2\pi^2 \text{i} g_s^2 \mathcal{I}_1(\omega).} \quad (2)$$

Only after the native calculation is frozen, the BRY coupling dictionary gives [1]

$$\boxed{-\text{i} \mathcal{A}_{1 \rightarrow 1}^{(1)}(it) = \frac{1}{2} \mathcal{I}_1(it).} \quad (3)$$

The external Liouville weight and internal weights are

$$d = 1 + \frac{\omega^2}{4}, \quad h(P) = 1 + P^2. \quad (4)$$

For $\omega = it$ with $0 < t < 1$, no DOZZ pole crosses a positive-real internal-momentum contour. The moduli integral is directly convergent: no physical-domain subtraction or crossed-pole residue enters the fifty-point dataset.

For the torus three-point calculation, the native and stripped normalizations used below are

$$\mathcal{A}_{1,3}^{\text{ws}} = 2\pi^2 \text{i} g_s^3 \mathcal{I}_{1,3}, \quad F_1^{\text{ws}} = \frac{\mathcal{I}_{1,3}}{4\pi}. \quad (5)$$

3 Current fifty-point $1 \rightarrow 1$ hybrid-recursion design

The punctured torus is split at $\epsilon = 0.15$. For the nearest lattice representative satisfying

$$|z_{\text{local}}| < 2\pi\epsilon, \quad (6)$$

the OPE channel uses

$$q = \exp(2\pi \text{i} \tau), \quad v = \exp(-\text{i} z_{\text{local}}) - 1, \quad (7)$$

and fixed-weight central-charge recursion through orders $(N_q, N_v) = (3, 8)$. Its flat-frame primary factor remains

$$q^{h_{\text{loop}} - 25/24} v^{h_{\text{OPE}}} [2 \sin(z_{\text{local}}/2)]^{-2d}. \quad (8)$$

The recursion changes only the descendant-coefficient generator and introduces no additional Weyl factor.

Away from collision, the necklace channel uses simultaneous internal-weight recursion through orders $(6, 3)$. Resonant terms at $c = 25$ are defined by a three-point Richardson limit in $c \rightarrow 25$ with production regulator $\delta c = 0.04$, and by a symmetric generic-weight limit at confluent poles with $\delta h = 10^{-3}$. Every coefficient tensor is independently audited at tolerance 10^{-7} ; failed nodes use exact finite-level descendant sewing.

The common numerical design is:

- four scrambled Sobol replicas with $2^8 = 256$ four-dimensional points per cutoff;
- bulk cutoffs $\tau_2 = 3, 4, 6, 8$;
- four tail replicas with $2^7 = 128$ points per slice at $\tau_2 = 8, 10, 12, 16, 20$;
- the tail ansatz $f(\tau_2) = a_0 \tau_2^{-2} + a_1 \tau_2^{-5/3} + a_2 \tau_2^{-3}$;

- two 16-point Gauss–Legendre momentum integrals with $P_{\max} = 6$ and threshold-resolving map $P = 6u^2$;
- collision-disc radius 0.10, with the leading OPE contribution integrated analytically;
- the same four scramble labels at all fifty energies.

The hybrid evaluator agrees with a complete direct-descendant rerun at $t = 0.45$ to 2.07×10^{-13} in $\mathcal{I}_1$, or 1.80×10^{-11} relatively. OPE c -recursion coefficients through (q^4, v^5) agree with direct coefficients to 1.35×10^{-14} at generic weights and 3.6×10^{-12} at coincident internal weights.

4 Frozen fifty-point $1 \rightarrow 1$ dataset

The quoted errors are the standard errors of the four shared-scramble RQMC means. They do not include block-order, momentum, collision-disc, or cusp-tail model systematics. The two halves of the energy grid are displayed side by side in Table 1.

Table 1: Current frozen fifty-point native worldsheet dataset.

t	$\mathcal{I}_1(it)$	σ_{QMC}	t	$\mathcal{I}_1(it)$	σ_{QMC}
0.02	$-3.35971048 \times 10^{-5}$	1.62×10^{-6}	0.52	$-1.34500776 \times 10^{-2}$	7.96×10^{-4}
0.04	$-1.34062367 \times 10^{-4}$	6.47×10^{-6}	0.54	$-1.38822705 \times 10^{-2}$	8.30×10^{-4}
0.05	$-2.09091786 \times 10^{-4}$	1.01×10^{-5}	0.55	$-1.40756228 \times 10^{-2}$	8.46×10^{-4}
0.07	$-4.07841860 \times 10^{-4}$	1.98×10^{-5}	0.57	$-1.44134033 \times 10^{-2}$	8.75×10^{-4}
0.10	$-8.23862706 \times 10^{-4}$	4.02×10^{-5}	0.60	$-1.47879798 \times 10^{-2}$	9.10×10^{-4}
0.12	$-1.17600960 \times 10^{-3}$	5.76×10^{-5}	0.62	$-1.49424840 \times 10^{-2}$	9.26×10^{-4}
0.14	$-1.58428488 \times 10^{-3}$	7.80×10^{-5}	0.64	$-1.50155537 \times 10^{-2}$	9.37×10^{-4}
0.15	$-1.80833205 \times 10^{-3}$	8.93×10^{-5}	0.65	$-1.50202590 \times 10^{-2}$	9.40×10^{-4}
0.17	$-2.29374374 \times 10^{-3}$	1.14×10^{-4}	0.67	$-1.49636491 \times 10^{-2}$	9.40×10^{-4}
0.20	$-3.10697282 \times 10^{-3}$	1.56×10^{-4}	0.70	$-1.47074092 \times 10^{-2}$	9.26×10^{-4}
0.22	$-3.69904326 \times 10^{-3}$	1.87×10^{-4}	0.72	$-1.44181503 \times 10^{-2}$	9.07×10^{-4}
0.24	$-4.32522823 \times 10^{-3}$	2.21×10^{-4}	0.74	$-1.40314033 \times 10^{-2}$	8.78×10^{-4}
0.25	$-4.64947974 \times 10^{-3}$	2.39×10^{-4}	0.75	$-1.38008957 \times 10^{-2}$	8.60×10^{-4}
0.27	$-5.31691750 \times 10^{-3}$	2.75×10^{-4}	0.77	$-1.32647535 \times 10^{-2}$	8.17×10^{-4}
0.30	$-6.35490981 \times 10^{-3}$	3.34×10^{-4}	0.80	$-1.22712081 \times 10^{-2}$	7.35×10^{-4}
0.32	$-7.06305135 \times 10^{-3}$	3.75×10^{-4}	0.82	$-1.14822848 \times 10^{-2}$	6.68×10^{-4}
0.34	$-7.77718831 \times 10^{-3}$	4.17×10^{-4}	0.84	$-1.05925320 \times 10^{-2}$	5.93×10^{-4}
0.35	$-8.13461085 \times 10^{-3}$	4.39×10^{-4}	0.85	$-1.01100702 \times 10^{-2}$	5.52×10^{-4}
0.37	$-8.84631647 \times 10^{-3}$	4.82×10^{-4}	0.87	$-9.07045177 \times 10^{-3}$	4.65×10^{-4}
0.40	$-9.89432405 \times 10^{-3}$	5.48×10^{-4}	0.90	$-7.32536319 \times 10^{-3}$	3.27×10^{-4}
0.42	$-1.05707747 \times 10^{-2}$	5.92×10^{-4}	0.92	$-6.04013651 \times 10^{-3}$	2.35×10^{-4}
0.44	$-1.12220612 \times 10^{-2}$	6.36×10^{-4}	0.94	$-4.66456393 \times 10^{-3}$	1.48×10^{-4}
0.45	$-1.15362905 \times 10^{-2}$	6.57×10^{-4}	0.95	$-3.94632373 \times 10^{-3}$	1.09×10^{-4}
0.47	$-1.21379894 \times 10^{-2}$	6.99×10^{-4}	0.97	$-2.43769807 \times 10^{-3}$	4.36×10^{-5}
0.50	$-1.29619911 \times 10^{-2}$	7.59×10^{-4}	0.99	$-8.36996171 \times 10^{-4}$	5.13×10^{-6}

The canonical manifest is

`results/genus1_two_point_worldsheet/imaginary_hybrid_hc_t_scan10_n256_v2/
worldsheet_scan_manifest.json.`

It records the fifty source paths, hashes, native values, errors, four replicas, cutoff sequence, tail slices, and common numerical design.

5 Current fifty-point fit and analytic comparison

Only after freezing the native data, all fifty values are mapped through (3) and fitted to the BRY ansatz

$$-i\mathcal{A}_{1\rightarrow 1}^{(1)}(it) = \frac{-at^2 + 2bt^4 - ct^5}{24}. \quad (9)$$

The RQMC-error-weighted fit is

$$a = 1.0016588 \pm 0.0565665, \quad b = 1.0053852 \pm 0.0666207, \quad c = 1.0093253 \pm 0.0764546. \quad (10)$$

The errors are obtained by fitting each of the four complete shared-scramble curves. The coefficients are strongly correlated. Imposing the analytic shape $a = b = c = \kappa$ gives

$$\kappa = 1.0082693 \pm 0.0240434. \quad (11)$$

The raw pointwise worldsheet/matrix-model ratios range from 0.99165 to 1.01474.

The matrix-model prediction in BRY notation is

$$-i\mathcal{A}_{1\rightarrow 1, \text{MM}}^{(1)}(it) = \frac{-t^2 + 2t^4 - t^5}{24}, \quad (a, b, c)_{\text{MM}} = (1, 1, 1). \quad (12)$$

BRY reported $(a, b, c) = (1.018, 1.028, 1.0344)$ [1]. The direct comparison is shown in Table 2.

Table 2: Current fifty-point fit compared with the exact matrix-model coefficients and the BRY numerical fit.

coefficient	matrix model	BRY	this work	shift from BRY
a	1	1.0180	1.001659 ± 0.056566	-1.61%
b	1	1.0280	1.005385 ± 0.066621	-2.20%
c	1	1.0344	1.009325 ± 0.076455	-2.42%

Relative to the exact matrix-model coefficients, BRY’s central values are high by 1.80%, 2.80%, and 3.44%, respectively. The current central values are high by 0.17%, 0.54%, and 0.93%. The current fit is closer to the matrix-model polynomial. There is no statistically significant disagreement with BRY: each BRY coefficient is only 0.29–0.34 marginal shared-scramble standard errors from the corresponding current value. With only four shared replicas and a strongly correlated covariance matrix, no joint significance is assigned.

At the exactly solvable resonance point $t = 2$, analytic continuation of (12) gives

$$-i\mathcal{A}_{1\rightarrow 1}^{(1)}(2i) = -\frac{1}{6}, \quad \mathcal{I}_1(2i) = -\frac{1}{3}. \quad (13)$$

The complete fit record, including the covariance matrix and replica fits, is

`results/genus1_two_point_worldsheet/imaginary_hybrid_hc_t_scan10_n256_v2/bry_postfreeze_fit_50point.json.`

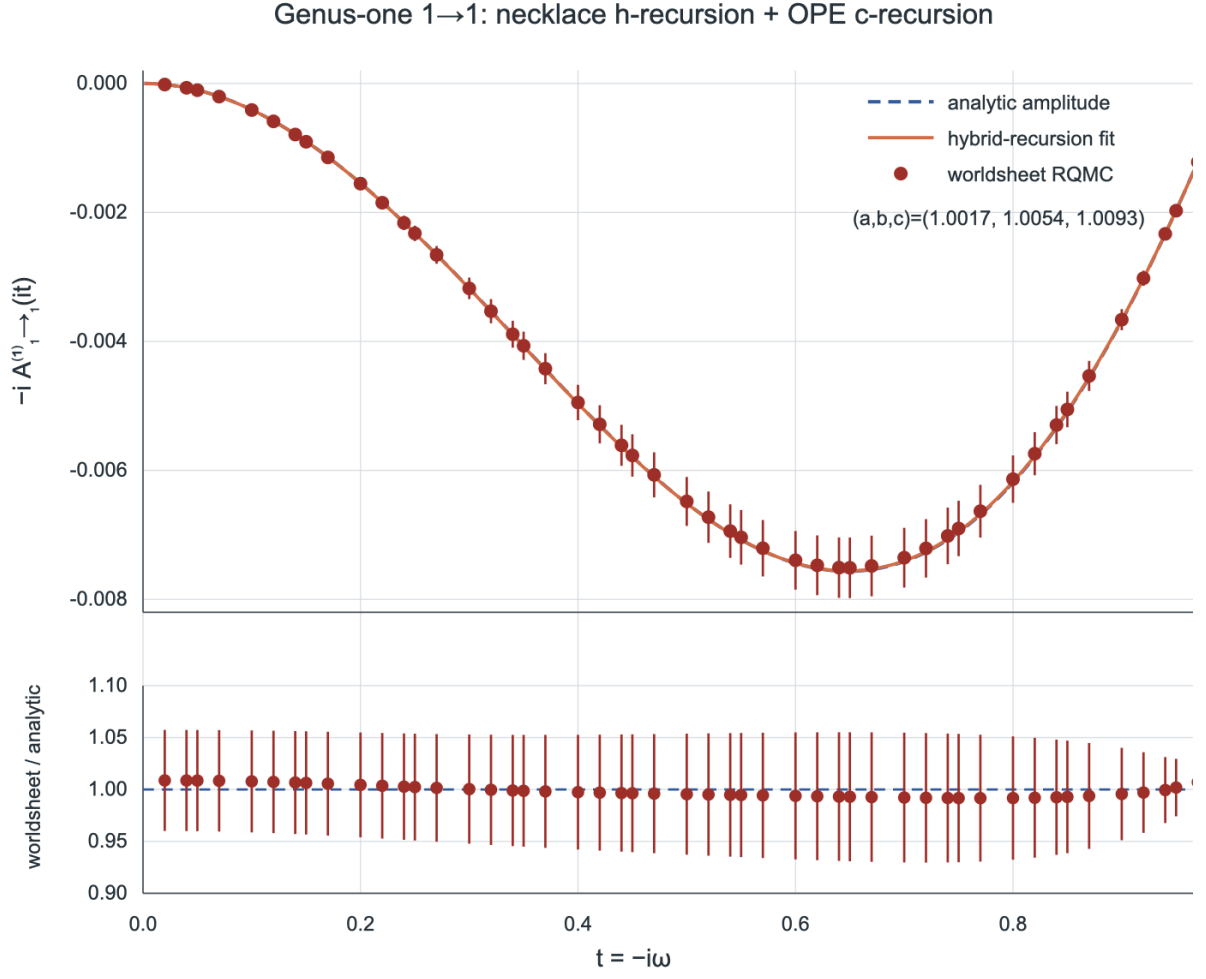


Figure 1: Current fifty-point genus-one $1 \rightarrow 1$ result. The upper panel shows the native worldsheet data after the BRY map, the hybrid-recursion fit, and the exact matrix-model prediction. The lower panel shows the pointwise ratio. Error bars are RQMC-only and are correlated through four shared scrambles.

6 Interpretation and limitations of the $1 \rightarrow 1$ result

The fifty-point curve is an audited numerical worldsheet result, not an exact evaluation of the generic torus integral. Its displayed errors measure the scatter of four shared RQMC replicas. They exclude possible systematic effects from finite block orders, momentum cutoff and quadrature, the leading-disc approximation, and the three-term cusp ansatz. The direct fallback and channel audits control the conformal-block implementation, while the full energy dependence gives a stringent global check.

The supported conclusions are:

1. the native worldsheet points reproduce the analytic genus-one shape across the convergent imaginary-energy strip;
2. the fitted normalization and three shape coefficients are consistent with the exact matrix-model value one;
3. the current central fit is closer to the exact polynomial than the BRY central fit, but the difference is not significant at the present RQMC precision;
4. a future precision claim should increase the number of independent shared scrambles and vary momentum, collision, block, and tail designs together;
5. extending through $t > 1$ requires the correct OPE-channel crossed-pole residue, or a necklace-channel implementation valid there.

No conclusion about higher torus amplitudes follows from the two-point calculation alone. The current three-point result below is an independent six-real-dimensional moduli integration and is not inferred from this fit.

7 Current torus three-point result: genus-one $1 \rightarrow 2$

7.1 Blind kinematics and integral

The symmetric imaginary-energy slice is

$$\omega = it, \quad \omega_1 = \omega_2 = \frac{it}{2}, \quad 0 < t < 1. \quad (14)$$

This is the convergent pole-free chamber: no DOZZ pole crosses a positive-real Liouville-momentum contour. Both outgoing positions are integrated over the full torus. In the necklace chart they are sorted along the cylinder and the three contracting gaps are

$$\log q_0 = i w_{(1)}, \quad \log q_1 = i(w_{(2)} - w_{(1)}), \quad \log q_2 = i(2\pi\tau - w_{(2)}). \quad (15)$$

Consequently $q_0 q_1 q_2 = \exp(2\pi i \tau)$ pointwise. The Liouville correlator is a threefold positive-momentum integral with measure $dQ_0 dQ_1 dQ_2 / \pi^3$ and three DOZZ coefficients. No matrix-model expression enters this kernel or the Cannon worker code.

7.2 Current h -recursion-dominant channel atlas

Most moduli points are evaluated in the necklace chart. Each cylinder variable is mapped to its elliptic nome $\hat{q}_a = E(q_a)$. The edge with largest $|\hat{q}_a|$ is expanded adaptively to the smallest order N satisfying

$$|\hat{q}_a|^{N+1} \leq 5 \times 10^{-5}, \quad (16)$$

with a floor of 3 and a prepared cap of 8; the other two edges are kept through order 3. The largest and second-largest necklace nomes encountered are 0.183754 and 0.049936. For every value of t , the selected orders are

$$N = 3 : 4076, \quad N = 4 : 19, \quad N = 5 : 1, \quad N = 6, 7, 8 : 0. \quad (17)$$

The simultaneous internal-weight recursion is regulated by $2F(25 + \epsilon) - F(25 + 2\epsilon)$ with $\epsilon = 0.025$. The common eta vacuum factor is cancelled algebraically against the Liouville primary vacuum factor before exponentiation. If the best necklace nome exceeds 0.30, or the second-best exceeds 0.07, a local pair- or comb-OPE chart is used instead. Those OPE blocks use fixed-weight c -recursion in their natural conformal frame, including the required frame factor. The largest OPE nome actually sampled is 0.067143, and no c -recursion collision occurs.

The current atlas therefore makes h -recursion the dominant method while reserving c -recursion for a small region. Per value of t , the recorded channel counts are 4096 necklace, 42 pair-OPE, and 336 comb-OPE evaluations. Their contributions to the final integral range over 70.98–72.21%, 14.81–19.38%, and 9.65–13.02%, respectively.

7.3 Frozen ten-point worldsheet scan

All ten points use one exact numerical design:

- four aligned scrambled Sobol replicas with $2^8 = 256$ six-dimensional points in every channel stratum;
- direct integration of the compact region through $\tau_2 = 8$ and an importance-sampled tail from $\tau_2 = 8$ to infinity, with proposal exponent $3/2$;
- three distinct but equivalent power–Legendre maps on $0 \leq Q \leq 5$, each of order 8;
- necklace orders $N_{\text{high}} = 8$ and $N_{\text{low}} = 3$, local OPE order 6, loop orders 4 and 2, patch radius 0.15, and triple-patch radius 0.10.

The assembled Cannon output contains nine new shards and reuses the existing $t = 0.75$ point only after exact-design and hash validation. Table 3 is the complete current result.

The curve rises from zero, has its largest sampled value at $t = 0.75$, and falls toward $t = 1$. A target-free quadratic through $t = 0.65, 0.75, 0.85$ gives

$$\boxed{t_{\text{peak}}^{\text{ws}} = 0.7164 \pm 0.0029} \quad (\text{aligned-scramble RQMC only}). \quad (18)$$

The relative RQMC errors decrease from 15.5% at $t = 0.05$ to 11.0% at $t = 0.95$. The direct tail contributes 6.98%–8.71%. These errors do not include a calibrated momentum-, block-, atlas-threshold-, or tail-proposal systematic.

Table 3: Current frozen target-blind torus three-point scan. The displayed error is the standard error of four aligned RQMC scrambles. The tail is integrated directly to infinity rather than fitted from fixed-height slices.

t	$-\text{Im } \mathcal{I}_{1,3}$	RQMC SE	tail fraction
0.05	0.0000281249	0.0000043727	8.71%
0.15	0.000729587	0.000113149	8.48%
0.25	0.00312010	0.000480503	8.18%
0.35	0.00759257	0.00115398	7.87%
0.45	0.01368245	0.00203629	7.60%
0.55	0.02008363	0.00289831	7.39%
0.65	0.02486229	0.00343811	7.23%
0.75	0.02581783	0.00337249	7.11%
0.85	0.02093083	0.00253755	7.03%
0.95	0.00885084	0.000973781	6.98%

7.4 Post-freeze comparison with the matrix model

The ten-point worldsheet manifest was first hash-frozen on Cannon. Its target-free local shape analysis was then independently frozen before the genus-one stripped matrix polynomial from Eq. (2.55) of Collier–Eberhardt–Rodriguez [2] was evaluated:

$$\hat{S}_{1,3} = -\frac{(\omega_1 + \omega_2 - i)(\omega_1 + \omega_2 - 2i)}{48} \left(\omega_1^2 - i\omega_1 + \omega_2^2 - i\omega_2 + 1 \right). \quad (19)$$

The stable three-point normalization already fixed at genus zero gives $F_0 = \omega\omega_1\omega_2$. CER expand their amplitudes using $g_s^{\text{CER}} = \sqrt{2}/\mu$, whereas the present expansion variable is $g_\mu = \mu^{-1} = 2\pi g_s$. The genus-one/tree ratio therefore contains $(g_s^{\text{CER}}/g_\mu)^2 = 2$, so that

$$F_1^{\text{MM}} = 2F_0\hat{S}_{1,3}. \quad (20)$$

On the equal imaginary ray this gives

$$F_1^{\text{MM}}(t) = -\frac{it^3(t-1)(t-2)(1+t-t^2/2)}{96}. \quad (21)$$

Equivalently, in the native ordinate shown in Table 4,

$$-\text{Im } \mathcal{I}_{1,3}^{\text{MM}}(t) = \frac{\pi t^3(t-1)(t-2)(1+t-t^2/2)}{24}. \quad (22)$$

All ten worldsheet points lie within one pointwise RQMC standard error of the matrix curve. The largest absolute pull is 0.838, and the ratios range from 0.8847 to 1.0186. The matrix curve peaks at $t \simeq 0.720$, consistent with Eq. (18). Since the ten estimates use only four aligned scrambles, the covariance matrix has low rank and no joint goodness-of-fit significance is assigned. No calibrated truncation systematic has been added to the displayed errors.

Table 4: Comparison evaluated only after the target-blind scan and shape analysis were frozen. Pulls use the pointwise RQMC standard errors only.

t	$-\text{Im } \mathcal{Z}_{1,3}^{\text{MM}}$	worldsheet / MM	RQMC pull
0.05	0.0000317891	0.8847	-0.838
0.15	0.000791100	0.9222	-0.544
0.25	0.00327169	0.9537	-0.315
0.35	0.00775727	0.9788	-0.143
0.45	0.01371520	0.9976	-0.016
0.55	0.01987684	1.0104	+0.071
0.65	0.02443801	1.0174	+0.123
0.75	0.02534664	1.0186	+0.140
0.85	0.02064459	1.0139	+0.113
0.95	0.00883076	1.0023	+0.021

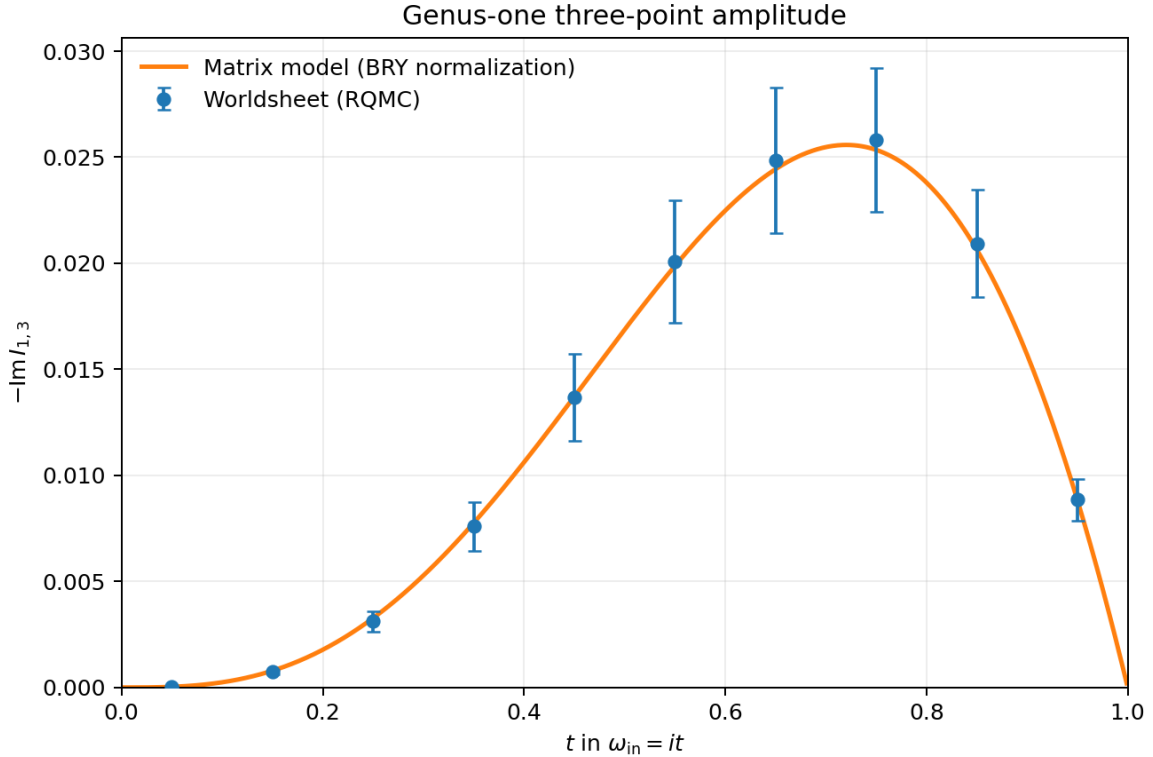


Figure 2: Current genus-one $1 \rightarrow 2$ result on the equal imaginary-energy ray. The blue worldsheet estimates are shown as unconnected points with RQMC error bars. The orange matrix-model curve is evaluated only after the worldsheet scan and target-free shape analysis were frozen.

8 Error budget and next gates

For every future torus $1 \rightarrow n$ production point, the following uncertainties should be retained separately:

1. adjacent momentum rules and changes of the high-momentum cutoff;
2. adjacent Virasoro-block orders and direct-sewing fallback counts;
3. moduli RQMC depth and additional independent scrambles;
4. agreement between overlapping necklace and OPE channels;
5. collision-disc radius and analytic leading-disc approximation;
6. direct-tail proposal exponent, tail boundary, and independent tail-stratum depth;
7. distance to the next Liouville-momentum contour wall and any crossed residue contribution.

The current $1 \rightarrow 1$ evidence identifies the analytic answer strongly in the controlled imaginary-energy chamber. The current $1 \rightarrow 2$ points agree with the BRY-normalized matrix curve at their RQMC precision, but the absence of a calibrated momentum-, block-, atlas-, and tail-proposal systematic prevents a precision discrepancy claim or a general torus-amplitude inference.

A Machine-readable provenance and implementation map

This appendix is a reviewer-oriented map of the code and frozen artifacts that support the torus $1 \rightarrow n$ claims. File names are relative to the repository root unless stated otherwise.

A.1 Trust boundary used in the blind calculations

The production flow is deliberately split into files with different authority:

1. a worldsheet kernel evaluates Liouville structure constants, timelike factors, conformal blocks, and moduli weights;
2. a target-blind scan produces raw worldsheet integrals and numerical diagnostics but contains no matrix-model polynomial;
3. the source JSON files are hashed and frozen in a manifest;
4. only a separate post-freeze fitter or comparison file applies the BRY convention map or evaluates the matrix-model function.

A.2 Torus $1 \rightarrow 1$ code path

Worldsheet kernel `plumbing/genus1_two_point_worldsheet.py` implements the native torus integrand, hybrid channel selection, and block backends.

RQMC production `plumbing/integrate_genus1_two_point_worldsheet.py` implements the

finite-cutoff moduli integration and cusp fit; `plumbing/run_genus1_two_point_imaginary_scan.py` runs the target-blind imaginary-ray scan.

Block audit `plumbing/genus1_two_point_hybrid_recursion_report.md` records the necklace h -recursion, OPE c -recursion, coincident-weight limits, and direct-descendant validation. The complete direct anchor is `plumbing/results/genus1_two_point_worldsheet/direct_current_t045_n256_v1.json`.

Later fit and plot `plumbing/fit_genus1_two_point_bry_scan.py` reads only frozen native artifacts before applying the BRY map; `plumbing/plot_genus1_two_point_hybrid_recursion.py` produces Figure 1.

The principal $1 \rightarrow 1$ artifacts are

- `plumbing/results/genus1_two_point_worldsheet/imaginary_hybrid_hc_t_scan10_n256_v2/worldsheet_scan_manifest.json`;
- `plumbing/results/genus1_two_point_worldsheet/imaginary_hybrid_hc_t_scan10_n256_v2/bry_postfreeze_fit_50point.json`;
- `plumbing/results/genus1_two_point_worldsheet/imaginary_hybrid_hc_t_scan10_n256_v2/genus1_two_point_hybrid_recursion.png`.

A.3 Torus $1 \rightarrow 2$ code path

Three-edge block `plumbing/torus_three_point_blocks.py` implements generic three-edge blocks and exact $c = 25$ sewing; `plumbing/torus_two_point_blocks.py` supplies the N -dimensional elliptic-nome re-expansion.

Worldsheet kernel and atlas

`plumbing/genus1_three_point_worldsheet.py` implements the target-free three-point kernel; `plumbing/genus1_three_point_channel_atlas.py` implements the h -dominant necklace/pair-OPE/comb-OPE atlas; `plumbing/smoke_genus1_three_point_channel_atlas.py` implements the stratified six-dimensional bulk and direct-tail RQMC.

Checks `plumbing/genus1_three_point_worldsheet_checks.py` checks the block and geometry; `plumbing/genus1_three_point_matrix_comparison_checks.py` checks only the separate post-freeze comparison formulas.

Cannon scan and assembly

`plumbing/run_genus1_three_point_hdominant_scan.py` fixes the exact ten-point design and reuse policy; `plumbing/genus1_three_point_hdominant_cannon.py` prepares and verifies the blind shards and writes the first worksheet freeze.

Blind analysis and later comparison

`plumbing/analyze_genus1_three_point_hdominant_scan.py` verifies every frozen hash, writes and hashes the target-free shape analysis, and only after that second freeze evaluates the BRY-normalized comparison.

The only current $1 \rightarrow 2$ result directory used in this note is

plumbing/results/genus1_three_point_worldsheet/hdominant_scan10_p8_h813_q030_007_n256_r4_v1/.

Its principal assembled artifacts are

- assembled/worldsheet_scan_manifest.json;
- assembled/worldsheet_freeze_manifest.json;
- assembled/worldsheet_shape_analysis_blind.json;
- assembled/worldsheet_analysis_freeze_manifest.json;
- assembled/matrix_comparison_after_blind_freeze.json;
- assembled/worldsheet_vs_matrix_bry.png.

A.4 Recommended faithfulness-review order

The shortest review path that checks every conceptual boundary is:

1. verify the sewing normalization and BRY coupling dictionary;
2. inspect the two-point hybrid-recursion audit and its direct anchor;
3. verify the fifty source hashes in the canonical manifest;
4. confirm by source inspection that the scan files contain no target polynomial;
5. reproduce the post-freeze fifty-point fit and plot;
6. inspect the three-edge block checks, channel atlas, and sorted-cylinder geometry;
7. verify all ten three-point shard hashes and the first worldsheet freeze;
8. reproduce the target-free shape analysis and verify its second freeze;
9. only then inspect the post-freeze BRY comparison.

Each amplitude section contains only its latest audited result. This note should next be extended by convergence variations of the current $1 \rightarrow 2$ design or by the first torus $1 \rightarrow 3$ integration.

References

- [1] B. Balthazar, V. A. Rodriguez, and X. Yin, “The $c = 1$ String Theory S-Matrix Revisited,” [arXiv:1705.07151](#).
- [2] S. Collier, L. Eberhardt, and V. A. Rodriguez, “ $c = 1$ Strings as a Matrix Integral,” [arXiv:2604.06301](#).